

Vikas Tiwari

Curriculum Vitae

I am a computational chemist exploring how molecular simulations, machine learning, and AI can be combined to uncover chemical mechanisms and accelerate discovery. My interests span machine-learned interatomic potentials, enhanced sampling, multiscale modeling, and data-driven drug discovery, with a focus on building predictive models for complex molecular and chemical systems.

Education

2021 – 2026

Ph.D. Computational Chemistry

Indian Institute of Technology Delhi

Thesis Advisor: Dr. Tarak Karmakar

CGPA: 9.50/10.0

Computational Modelling of Monolayer-Protected Metal Nanoclusters: Formation, Self-Assembly, and Catalytic Mechanisms

Monolayer-protected metal nanoclusters (MPCs) are a new class of materials with atomically precise structures, enabling tuneable physicochemical properties and a broad range of applications in catalysis, biomedicine, and materials science. This thesis employs a multiscale computational approach to understand key aspects of MPC behaviour, including their formation, self-assembly, interactions with biomolecules, and catalytic mechanisms.

2018 – 2020

M.Sc. Chemistry

Indian Institute of Technology Delhi

Thesis Advisor: Prof. Anil J. Elias

CGPA: 9.33/10.0

Highly E-Selective Olefination of Methyl Substituted N-Heteroarenes with Benzyl Amines using NaCl as Catalyst and Water as Solvent

Developed a sustainable, metal-free method for highly E-selective olefination using NaCl catalyst in aqueous medium. Demonstrated broad substrate scope and green synthesis potential through in-situ aldehyde formation.

2015 – 2018

B.Sc. Chemistry

B. K. Birla College of Commerce, Science and Arts, Kalyan, Mumbai

CGPA: 6.87/7.00

Scholarships and Awards

2024

Asian Young NanoScientist Presentation Award

2021

Prime Minister Research Fellowship

2020

CSIR-NET Junior Research Fellowship Award - All India Rank 28

2020

Graduate Aptitude Test in Engineering (GATE) - All India Rank 31

Research Skills

- Classical molecular dynamics simulations (Atomistic and coarse grained)
- Ab initio molecular dynamics simulations
- Enhanced sampling methods (Metadynamics, OPES, Umbrella sampling)
- Machine-learning potential-based simulations
- QM/MM molecular dynamics simulations
- Mechanistic investigation using static DFT calculations

Technical Skills

- **OS:** Linux, Windows, macOS
- **Software:** GROMACS, Gaussian, VMD, OVITO, PLUMED, CP2K, DeepMD-kit, LAMMPS, Quantum ESPRESSO
- **Programming:** Python (NumPy, Pandas, Matplotlib, ASE)
- **ML Frameworks:** PyTorch, LibTorch
- **Visualization:** Matplotlib, VMD, OVITO
- **Tools:** GitHub, LaTeX, Jupyter Notebooks